Ahmedabad University

Weekly Report 5

Disease Prediction System using Machine Learning

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Course: CSE 623 Machine Learning

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Week 5 – Baseline Model Development

Work Completed

This week, the first set of baseline machine learning models was implemented and trained on the prepared training dataset. The objective of this stage was to establish initial performance benchmarks for disease prediction using commonly used classification algorithms.

Two models were implemented: a Decision Tree Classifier and a Gaussian Naive Bayes model. Both models were trained on the symptom-feature dataset to learn patterns associating specific symptom combinations with particular diseases.

After training the models, the testing dataset was passed through each model to generate the first set of disease predictions. These predictions were then compared with the actual disease labels to compute initial accuracy scores.

The results obtained during this stage provide a baseline reference that can later be used to compare the performance of more advanced models.

Tools / Technologies Used

- Python
- Scikit-learn machine learning library

Challenges Faced

Ensuring that the models are properly trained and that the training and testing workflow produces valid predictions without data leakage.

Plan for Next Week

- Implement a Random Forest classifier as an advanced model.
- Compare its performance with the baseline models.